

MLOps feature engineering steps

1.  **Raw Data Ingestion (daily/hourly)**
 - Data is ingested into your data lake (e.g. S3) continuously.
2.  **Data Cleaning & Feature Engineering**
 - Use **Pandas** or **Dask** to clean up raw data and do feature engineering.
 - Save cleaned data back to S3 (or other persistent store).
3.  **Data Validation**
 - Run **TensorFlow Data Validation (TFDV)** on the cleaned data.
 - Detect schema violations, anomalies, and drift.
4.  **Data Transformation**
 - Apply **TensorFlow Transform (TFT)** to scale, normalize, and encode features.
 - Save the transformation graph so training and serving share the same logic.
5.  **Prepare Evaluation Data**
 - Split off a **hold-out slice** of the new data as an **eval set** for continuous monitoring.
6.  **Model Performance Evaluation**
 - Run **TFMA (TensorFlow Model Analysis)** on the eval set.
 - Check metrics (e.g. Accuracy, AUC, Precision) on new data slices.
7.  **Metrics Logging & Monitoring**
 - Log the metrics to a monitoring solution (e.g. MLflow, CloudWatch, Prometheus/Grafana) for time-series tracking.
 - Set up alerts if performance drops below a defined threshold.
8. **Automatic Retraining Trigger**
 - If metrics degrade (drift detected or KPI < threshold), trigger a **model retraining pipeline**.
 - Re-run training with up-to-date data and re-deploy the new model version.